AyuAware Today Section - Visual Design Specification

Overview

The Today section is the primary daily interface where users interact with their Digital Twin. It balances at-a-glance health metrics with conversational AI guidance, maintaining the "chat-first" philosophy while providing quick access to key data.

Design Principles

1. **Chat-First**: Conversation is the primary interaction mode, metrics are supporting context
2. **Glanceable**: Key metrics visible without scrolling
3. **Progressive Disclosure**: Details revealed on demand, not overwhelming upfront
4. **Persistent Access**: Chat input always reachable
5. **Contextual Intelligence**: Content adapts to time of day and user state

Color Palette

Background Colors

- **App Background**: `#F5F7F6` (very light gray-green)
- **Chat Background**: `#FFFFFF` (white for messages area)
- **Header Background**: `#FFFFFF` with subtle shadow
- **Metric Card Background**: `#FFFFFF`

Message Colors

- **Twin Message Background**: `#F8FAFB` (light teal-gray)
- **User Message Background**: `#E8F4F3` (soft teal)
- **System Message**: `#FFF9F0` (warm off-white for celebrations/milestones)

Accent Colors

- **Primary Teal**: `#4A9B8E`
- **Secondary Teal**: `#7AB8AC`
- **Warm Gold**: `#F8C471` (for highlights/celebrations)
- **Alert Orange**: `#E89B7F` (for important notices)
- **Calm Blue**: `#A8C8E1` (for sleep/rest contexts)

Status Colors

- **CHI - Good**: `#A8E6A3` (soft green)
- **CHI - Moderate**: `#F8C471` (warm gold)
- **CHI - Low**: `#E89B7F` (warm coral)
- **Vata**: `#7AB8AC` (air/ether - teal)

- **Pitta**: `#E89B7F` (fire - coral)
- **Kapha**: `#A3B899` (earth - sage green)

Text Colors

- **Primary Text**: `#1A2B26` (deep green-black)
- **Secondary Text**: `#5A6B66` (medium gray-green)
- **Tertiary Text**: `#A8B5B1` (light gray)
- **On Teal**: `#FFFFFF` (white)

Typography

Headers

- **Page Title** (Today): SF Pro/Inter, 28px, weight 700, letter-spacing -0.5px
- **Section Title**: 20px, weight 600, letter-spacing -0.3px
- **Card Title**: 16px, weight 600

Body Text

- **Message Text**: 16px, weight 400, line-height 24px
- **Metric Label**: 13px, weight 600, letter-spacing 0.3px, uppercase
- **Metric Value**: 24px, weight 700, letter-spacing -0.5px
- **Metric Subtitle**: 13px, weight 500

UI Elements

- **Button Text**: 15px, weight 600
- **Chip Text**: 14px, weight 500
- **Input Placeholder**: 16px, weight 400, italic
- **Timestamp**: 12px, weight 400

Layout Structure

Screen Dimensions (iPhone 14 Pro as baseline)

- Screen Width: 393px
- Screen Height: 852px
- Safe Area Top: 59px (includes status bar)
- Safe Area Bottom: 34px (home indicator)

Grid System

- Horizontal Margins: 16px
- Content Max Width: 361px (393 - 32px margins)
- Column Grid: 12 columns with 8px gutters

Spacing Scale (8px base)

- **4px**: Micro spacing (icon + text)
- **8px**: Tight spacing (related elements)
- **12px**: Default spacing (message padding)

- **16px**: Standard spacing (section margins)
- **24px**: Large spacing (section breaks)
- **32px**: XL spacing (major sections)

Component Specifications

1. Header Bar

```
┌──────────────────────────────────────────────────────────┐
│  ≡  Today                Sarah          🔔  ⚙️          │
│     ──────────────────────────────────────────────       │
└──────────────────────────────────────────────────────────┘
```

Specifications:

- Height: 56px + safe area top
- Background: `#FFFFFF`
- Border bottom: 1px solid `#E0EBE8`
- Shadow: `0 1px 3px rgba(0,0,0,0.04)`
- Z-index: 100 (above content)
- Position: Sticky at top

Left Side (Menu Icon):

- Icon: Hamburger menu (3 lines)
- Size: 24x24px
- Color: `#1A2B26`
- Tap area: 44x44px
- Position: 16px from left edge

Center (Title + Name):

- "Today": 17px, weight 600, color `#1A2B26`
- User name: 15px, weight 400, color `#5A6B66`
- Layout: Vertical stack, centered
- Max width: 200px (truncate with ellipsis)

Right Side (Actions):

- Notification bell: 22x22px
- Settings gear: 22x22px
- Color: `#1A2B26`
- Spacing: 16px between icons
- Tap area: 44x44px each
- Position: 16px from right edge

Badge (on notification):

- Size: 8x8px circle
- Background: `#E89B7F`
- Position: Top-right of icon, offset -2px
- Pulse animation if unread

2. Time-Based Greeting

```
┌──────────────────────────────────────────────────────────┐
│  Good morning, Sarah 🌅                                  │
│  Friday, August 11, 2023                                 │
└──────────────────────────────────────────────────────────┘
```

Specifications:

- Background: Subtle gradient from `#FFFFFF` to `#F5F7F6`
- Padding: 16px horizontal, 20px vertical
- Border bottom: 1px solid `#E0EBE8`

Greeting Text:

- Font size: 20px
- Weight: 600
- Color: `#1A2B26`
- Includes contextual emoji
- Line height: 28px

Date Text:

- Font size: 14px
- Weight: 400
- Color: `#5A6B66`
- Margin top: 4px

Time-Based Variations:

Morning (5 AM - 12 PM):

- Text: "Good morning, [Name] 🌅"
- Gradient: Warm tint (subtle yellow)

Afternoon (12 PM - 5 PM):

- Text: "Good afternoon, [Name] ☀️"
- Gradient: Bright tint (subtle orange)

Evening (5 PM - 9 PM):

- Text: "Good evening, [Name] 🌆"
- Gradient: Cool tint (subtle purple)

Night (9 PM - 5 AM):

- Text: "Good evening, [Name] 🌙"
- Gradient: Dark blue tint

3. Compact Metric Cards Row

```
┌──────────────────────────────────────────────────────────┐
│  ┌──────────┐  ┌──────────┐  ┌──────────┐               │
│  │ CHI      │  │ NSL      │  │ Exposure │               │
```

```
│   │    72%     │   │    82%     │   │            │        │
│   │  Kapha↑     │   │  Vata↑     │   │  ☀️ 🌡️    │        │
│   └─────────────┘   └────────────┘   └────────────┘        │
│                                                             │
│   Tap any card to explore ↑                                 │
└─────────────────────────────────────────────────────────────┘
```

Container Specifications:

- Background: `#F5F7F6` (matches app background)
- Padding: 16px all sides
- Border bottom: 1px solid `#E0EBE8`

Card Grid:

- Layout: 3 columns, equal width
- Gap: 8px between cards
- Horizontal scroll if needed (for additional metrics)

Individual Card Specifications:

Card Container:

- Background: `#FFFFFF`
- Border: 1.5px solid `#E0EBE8`
- Border radius: 12px
- Padding: 12px
- Min height: 100px
- Shadow: `0 1px 3px rgba(74, 155, 142, 0.06)`
- Tap area: Full card

Tap State:

- Transform: scale(0.97)
- Duration: 100ms
- Border: 1.5px solid `#4A9B8E`
- Shadow: `0 2px 6px rgba(74, 155, 142, 0.12)`

Card Content Layout:

Label (Top):

- Font size: 13px
- Weight: 600
- Color: `#5A6B66`
- Letter spacing: 0.3px
- Text transform: uppercase
- Margin bottom: 8px

Value (Middle):

- Font size: 32px
- Weight: 700
- Color: `#1A2B26`
- Line height: 36px
- Letter spacing: -1px

Status/Subtitle (Bottom):

- Font size: 12px
- Weight: 500
- Color: varies by status
- Margin top: 4px
- May include icon (↑↓→) 16x16px

Status Colors by Value:

CHI Card:

- 80-100%: Text `#4A9B8E`, Icon 🟢
- 60-79%: Text `#F8C471`, Icon 🟡
- 0-59%: Text `#E89B7F`, Icon 🟠

NSL Card:

- Uses tri-color circular progress (Vata/Pitta/Kapha)
- Dominant dosha shown in subtitle with ↑

Exposure Card:

- Weather icon: 32x32px
- Temperature icon: 24x24px
- No percentage value
- Shows current weather context

Helper Text:

- Font size: 13px
- Weight: 400
- Color: `#A8B5B1`
- Text align: center
- Margin top: 12px
- Includes small up arrow ↑ 14x14px

4. Chat Messages Area

Container:

- Background: `#FFFFFF`
- Padding: 16px horizontal
- Spacing: 12px between message groups
- Minimum height: calc(100vh - header - metrics - input - safe areas)

Twin Message (Left-aligned)

```
┌──────────────────────────────────────────────────────────┐
│                                                          │
│  [🪷]                                                    │
│   ┌──────────────────────────────────────────┐           │
│   │  I've analyzed your overnight data.      │           │
│   │  Here's what I see:                      │           │
```

```
Your recovery looks good overall
with steady HRV and balanced heart
rate. However, a slight Kapha
imbalance is showing up in lower
activity levels.

7:30 AM
```

Avatar:

- Size: 32x32px
- Border radius: 50% (circle)
- Position: Top-left, aligned with first line of message
- Image: Lotus icon with subtle glow
- Margin right: 8px

Message Bubble:

- Background: `#F8FAFB`
- Border radius: 16px (all corners except top-left: 4px - chat tail)
- Padding: 12px 16px
- Max width: 75% of container (271px on iPhone 14 Pro)
- Margin left: 40px (avatar + spacing)
- Box shadow: `0 1px 2px rgba(0,0,0,0.04)`

Text:

- Font size: 16px
- Weight: 400
- Line height: 24px
- Color: `#1A2B26`
- Text align: left
- Word wrap: break-word

Emphasis in Text:

- Bold text: weight 600 (not 700)
- Italic: preserve style
- Links: color `#4A9B8E`, underline on tap

Timestamp:

- Font size: 12px
- Weight: 400
- Color: `#A8B5B1`
- Position: Below bubble, left-aligned
- Margin top: 4px
- Margin left: 40px (aligned with message start)

User Message (Right-aligned)

```
Why is my CHI lower than
yesterday?

                    7:31 AM
```

Message Bubble:

- Background: `#E8F4F3`
- Border radius: 16px (all corners except top-right: 4px)
- Padding: 12px 16px
- Max width: 75% of container
- Margin right: 0
- Align: right
- Box shadow: `0 1px 2px rgba(0,0,0,0.04)`

Text:

- Same specs as Twin message
- Color: `#1A2B26`

Timestamp:

- Position: Below bubble, right-aligned
- Other specs same as Twin timestamp

System Message (Centered)

```
———————— Morning check-in ————————
completed ✓
7:30 AM
```

Specifications:

- Background: None (transparent)
- Text align: center
- Font size: 13px
- Weight: 500
- Color: `#5A6B66`
- Padding: 12px 0
- Border top/bottom: 1px solid `#E0EBE8` (optional, for separation)

Line Decoration:

- Width: 32px each side
- Height: 1px
- Color: `#E0EBE8`
- Position: Vertically centered with text

5. Rich Message Components

Metric Card in Message

```
┌──────────────────────────────────────────────────────────┐
│  [🪷]                                                    │
│  ┌──────────────────────────────────────────────────┐    │
│  │  Your CHI overview:                              │    │
│  │                                                  │    │
│  │  ╔═══════════════════════════════════════════╗   │    │
│  │  ║                                           ╕   │    │
│  │  │   ┌──────────────────┐                    │   │    │
│  │  │   │                  │                    │   │    │
│  │  │   │   72%            │                    │   │    │
│  │  │   │  [Chart]         │                    │   │    │
│  │  │   │                  │                    │   │    │
│  │  │   └──────────────────┘                    │   │    │
│  │  │                                           │   │    │
│  │  │  Chi feels steady with                    │   │    │
│  │  │  good recovery reflected                  │   │    │
│  │  │  in your HRV                              │   │    │
│  │  │                                           │   │    │
│  │  │  [Explore CHI details →]                  │   │    │
│  │  ╒                                           ╕   │    │
│  │  ╚═══════════════════════════════════════════╝   │    │
│  │                                                  │    │
│  └──────────────────────────────────────────────────┘    │
│                                                          │
│  7:30 AM                                                 │
└──────────────────────────────────────────────────────────┘
```

Embedded Card:

- Background: `#FFFFFF`
- Border: 1.5px solid `#E0EBE8`
- Border radius: 10px
- Padding: 16px
- Margin: 12px 0 (within message bubble)
- Shadow: `0 1px 4px rgba(74, 155, 142, 0.08)`

Chart Visualization:

- Size: 120x120px
- Type: Circular progress (donut style)
- Colors: Based on metric type
- Animation: Animate fill on appearance (500ms ease-out)
- Center text: Value in large type

Description Text:

- Font size: 15px
- Weight: 400
- Line height: 22px
- Color: `#5A6B66`

- Margin top: 12px from chart

Action Button:

- Style: Text button with arrow
- Font size: 15px
- Weight: 600
- Color: `#4A9B8E`
- Padding: 8px 12px
- Border radius: 6px
- Background: `#F0F9F7` on tap
- Margin top: 12px
- Align: right

6. Quick Action Grid

```
┌──────────────────────────────────────────────────────────┐
│  [🪷]                                                    │
│   ┌──────────────────────────────────────────────┐       │
│   │  What would you like to do?                  │       │
│   │                                              │       │
│   │   ┌──────────┐   ┌──────────┐               │       │
│   │   │    🥗    │   │    🏃    │               │       │
│   │   │ Log      │   │ Start    │               │       │
│   │   │ Breakfast│   │ Workout  │               │       │
│   │   └──────────┘   └──────────┘               │       │
│   │                                              │       │
│   │   ┌──────────┐   ┌──────────┐               │       │
│   │   │    🧘    │   │    📋    │               │       │
│   │   │ Meditate │   │ View     │               │       │
│   │   │          │   │ Protocols│               │       │
│   │   └──────────┘   └──────────┘               │       │
│   └──────────────────────────────────────────────┘       │
│  7:33 AM                                                 │
└──────────────────────────────────────────────────────────┘
```

Container:

- Embedded within Twin message bubble
- Uses full width of bubble
- Padding: 12px 0 (top and bottom within bubble)

Question Text:

- Font size: 16px
- Weight: 500
- Color: `#1A2B26`
- Margin bottom: 12px

Grid Layout:

- Columns: 2
- Rows: 2
- Gap: 10px between items
- Equal width items

Action Card:

- Background: `#FFFFFF`
- Border: 1.5px solid `#E0EBE8`
- Border radius: 12px
- Padding: 16px 12px
- Aspect ratio: ~1:1 (square-ish)
- Tap area: Full card

Tap State:

- Transform: scale(0.95)
- Duration: 100ms
- Background: `#F8FAFB`
- Border: 1.5px solid `#4A9B8E`

Icon:

- Size: 32x32px
- Position: Center-top
- Margin bottom: 8px
- Can be emoji or custom icon

Label:

- Font size: 14px
- Weight: 600
- Color: `#1A2B26`
- Text align: center
- Line height: 18px
- Max lines: 2
- Overflow: ellipsis

7. Suggested Questions

```
┌──────────────────────────────────────────────────┐
│  💭 Why is my CHI low today?                     │
│                                                  │
│  💭 What's the best time for my practice?        │
│                                                  │
│  💭 How is the moon affecting me?                │
│                                                  │
│  💭 Why did you suggest lemon water?             │
└──────────────────────────────────────────────────┘
```

Container:

- Background: `#FFFFFF`

- Padding: 16px horizontal, 12px vertical
- Border top: 1px solid `#E0EBE8`
- Position: Above input bar

Section Label (Optional):

- Text: "Try asking..."
- Font size: 13px
- Weight: 500
- Color: `#A8B5B1`
- Margin bottom: 12px

Question Chip:

- Background: `#F8FAFB`
- Border: 1px solid `#E0EBE8`
- Border radius: 20px (full pill)
- Padding: 10px 16px
- Margin: 6px 0
- Full width
- Text align: left

Chip Text:

- Font size: 14px
- Weight: 500
- Color: `#1A2B26`
- Line height: 20px

Icon:

- Emoji: 💭 (thought bubble)
- Size: 16x16px
- Position: Left, 12px from edge
- Margin right: 8px from text

Tap State:

- Background: `#E8F4F3`
- Border: 1px solid `#4A9B8E`
- Transform: scale(0.98)
- Duration: 100ms

Hover/Press Animation:

- Slight scale down
- Color shift to show interactivity

Loading State (after tap):

- Show typing indicator immediately
- Chip fades out (300ms)
- Question appears as user message

8. Chat Input Bar (Sticky)

```
┌──────────────────────────────────────────────────────────────┐
```

```
│  📷   Ask your twin anything...      🎤   ➤     │
└──────────────────────────────────────────────────┘
```

Container:

- Background: `#FFFFFF`
- Height: 56px + safe area bottom
- Padding: 8px 16px
- Border top: 1px solid `#E0EBE8`
- Shadow: `0 -2px 8px rgba(0,0,0,0.04)`
- Position: Fixed at bottom
- Z-index: 99

Layout:

- Display: flex
- Align items: center
- Gap: 12px between elements

Camera/Image Button (Left):

- Icon: Camera (📷 or custom)
- Size: 24x24px
- Color: `#5A6B66`
- Tap area: 44x44px
- Border radius: 50%
- Background on tap: `#F5F7F6`

Text Input:

- Background: `#F5F7F6`
- Border: 1px solid `#E0EBE8`
- Border radius: 20px (pill shape)
- Padding: 10px 16px
- Flex: 1 (takes remaining space)
- Font size: 16px
- Weight: 400
- Color: `#1A2B26`
- Placeholder color: `#A8B5B1`
- Min height: 40px
- Max height: 120px (expands with content)

Focus State:

- Border: 1px solid `#4A9B8E`
- Background: `#FFFFFF`
- Box shadow: `0 0 0 3px rgba(74, 155, 142, 0.1)`

Voice Button (Right-Middle):

- Icon: Microphone (🎤 or custom)
- Size: 24x24px
- Color: `#5A6B66`
- Tap area: 44x44px
- Border radius: 50%

- **Recording state**:
  - Color: `#E89B7F`
  - Pulsing animation
  - Show waveform visualization

Send Button (Right):

- Icon: Arrow (➤ or paper plane)
- Size: 24x24px
- Background: `#4A9B8E`
- Color: `#FFFFFF`
- Border radius: 50%
- Width/Height: 40x40px
- **Disabled state**:
  - Background: `#E0EBE8`
  - Color: `#A8B5B1`
  - No tap interaction

Tap State (Send):

- Transform: scale(0.9)
- Duration: 100ms
- Shadow: `0 2px 4px rgba(74, 155, 142, 0.3)`

9. Typing Indicator

```
┌──────────────────────────────────────────────────────┐
│  [ 🪷 ]                                              │
│  ┌──────────────────────────────────┐                │
│  │  ● ● ●                           │                │
└──────────────────────────────────────────────────────┘
```

Container:

- Same styling as Twin message bubble
- Min height: 44px
- Width: 60px
- Padding: 12px 20px

Dots:

- Size: 8x8px circles
- Color: `#7AB8AC`
- Spacing: 6px between dots
- Animation: Wave pattern
  - Duration: 1.4s
  - Ease: ease-in-out
  - Loop: infinite
  - Each dot animates with 0.2s delay

Animation Keyframes:

```
0%, 60%, 100%: opacity 0.3, translateY(0)
```

```
30%: opacity 1, translateY(-8px)
```

10. Empty State (No Messages)

```
┌──────────────────────────────────────────────────────────┐
│                                                          │
│                                                          │
│                    ┌──────────────┐                      │
│                    │              │                      │
│                    │   🪷 [Lotus] │                      │
│                    │              │                      │
│                    └──────────────┘                      │
│                                                          │
│              Hi Sarah! I'm your Twin.                    │
│                                                          │
│           I'm here to help you understand                │
│           your body and maintain balance.                │
│                                                          │
│                What would you like to                    │
│                explore today?                            │
│                                                          │
└──────────────────────────────────────────────────────────┘
```

Container:

- Full height of chat area
- Vertical align: center
- Text align: center
- Padding: 40px 32px

Lotus Icon:

- Size: 120x120px
- Subtle glow animation
- Margin bottom: 24px

Welcome Text:

- Font size: 18px
- Weight: 600
- Color: `#1A2B26`
- Line height: 26px
- Margin bottom: 12px

Description:

- Font size: 16px
- Weight: 400
- Color: `#5A6B66`
- Line height: 24px
- Max width: 280px

- Margin: 0 auto

11. Date Selector (Historical View)

```
┌──────────────────────────────────────────────────┐
│          ←    August 11, Monday    →              │
└──────────────────────────────────────────────────┘
```

When in Historical Mode:

- Appears below greeting, above metrics
- Background: `#FFF9F0` (warm tint to indicate past)
- Padding: 12px 16px
- Border: 1px solid `#F8C471`
- Border radius: 8px
- Margin: 12px 16px

Date Text:

- Font size: 16px
- Weight: 600
- Color: `#1A2B26`
- Text align: center

Arrow Buttons:

- Size: 24x24px
- Color: `#4A9B8E`
- Tap area: 44x44px
- Position: Left and right sides
- **Disabled state**: Color `#E0EBE8`

Today Badge:

- If viewing today, show "Today" badge
- Background: `#4A9B8E`
- Color: `#FFFFFF`
- Font size: 12px
- Weight: 600
- Padding: 4px 12px
- Border radius: 12px
- Position: Right side of date

12. Pull-to-Refresh

Visual Feedback:

- Pull distance threshold: 80px
- Spinner appears at 40px pull
- Release to refresh at 80px

Spinner:

- Size: 24x24px
- Color: `#4A9B8E`

- Style: iOS-style activity indicator
- Position: Top center of chat area
- Animation: Rotation, 1s duration

Refresh Animation:

- Haptic feedback on release
- Spinner continues for minimum 500ms
- Fade out on completion

Interactions & Animations

Message Appearance

- **Entry**: Fade + slide from bottom
- Duration: 300ms
- Easing: ease-out
- Stagger: 100ms between messages if multiple

Metric Card Tap

On Tap:

- Haptic feedback (light impact)
- Scale: 0.97
- Duration: 100ms
- Navigate to detailed view

Transition:

- Hero animation from card to full screen
- Duration: 400ms
- Easing: ease-in-out

Quick Action Tap

On Tap:

- Haptic feedback (light impact)
- Scale: 0.95
- Duration: 100ms
- Show confirmation or next step

Suggested Question Tap

On Tap:

- Chip fades out (200ms)
- Question appears as user message immediately
- Twin typing indicator appears
- Smooth scroll to bottom

Send Message

On Send:

- Haptic feedback (medium impact)
- Message appears instantly as user message
- Input clears immediately
- Twin typing indicator after 300ms delay
- Auto-scroll to bottom

Image Upload

On Select:

- Image preview appears in input bar
- Thumbnail: 40x40px with close button
- Can add text with image
- Send when ready

Voice Recording

On Hold:

- Haptic feedback (heavy impact) on start
- Button pulses
- Waveform visualization appears
- Timer shows duration
- Release to send, swipe left to cancel

Responsive Behavior

iPhone SE (375px width)

- Metric cards: 2 visible, 3rd scrollable
- Message max width: 70% (262px)
- Font sizes: Same (don't reduce)
- Padding: 12px (reduce from 16px)

iPhone 14 Pro (393px width)

- Standard specs apply
- 3 metric cards fit comfortably

iPhone Pro Max (428px width)

- Metric cards: 3 full width + spacing
- Message max width: 75% (321px)
- More breathing room overall

iPad (768px+ width)

- Max container width: 600px
- Center content
- Larger side margins (32px)
- Message max width: 70% of container

Landscape Mode

- Reduce header height: 48px

- Combine greeting + date into header
- More horizontal space for messages
- Consider 4 quick action cards instead of 2x2

Accessibility

WCAG AA Compliance

- All text contrast: minimum 4.5:1
- Touch targets: minimum 44x44px
- Focus indicators: 2px solid `#4A9B8E`

VoiceOver Support

- All interactive elements labeled
- Message roles: "message from Twin" / "message from you"
- Metric cards: "CHI at 72%, Kapha imbalance, button"
- Input: "Message input field"
- Actions announced on tap

Dynamic Type Support

- Respect user's text size preferences
- Layout adjusts for larger text
- Minimum: -2 size categories
- Maximum: +5 size categories

Reduced Motion

- Disable animations if user prefers
- Keep essential transitions only
- No parallax or complex motion

Color Blind Modes

- Don't rely solely on color for status
- Use icons + text labels
- High contrast mode available

Error States

Network Error

```
┌──────────────────────────────────────────────────────────┐
│   [🪷]                                                    │
│   ┌──────────────────────────────────────┐                │
│   │  ⚠️ Couldn't send that message        │                │
│   │                                      │                │
│   │  Check your connection and           │                │
│   │  try again.                          │                │
```

```
│   │                                   │       │
│   │  [Retry]  [Dismiss]               │       │
│   └───────────────────────────────────┘       │
└───────────────────────────────────────────────┘
```

Error Message:

- Background: `#FFF5F5` (light red tint)
- Border: 1px solid `#E89B7F`
- Same bubble styling otherwise

Icon:

- Warning triangle: 20x20px
- Color: `#E89B7F`

Buttons:

- Retry: Primary style (`#4A9B8E`)
- Dismiss: Secondary style (outlined)

Failed Metric Load

```
┌──────────┐
│  CHI     │
│  --      │
│Tap to    │
│retry     │
└──────────┘
```

Failed State:

- Value shows: `--`
- Background: `#FFF5F5`
- Border: 1px solid `#E89B7F`
- Subtitle: "Tap to retry"
- Icon: Refresh symbol

API Timeout

- Show gentle message after 10s
- "Twin is taking longer than usual..."
- Option to cancel and try again
- Keep typing indicator active

Performance Considerations

Lazy Loading

- Messages: Load 20 at a time
- Infinite scroll upward for history
- Images: Load on visibility
- Metrics: Fetch on screen appearance

Optimistic Updates

- Messages appear instantly
- Show pending state (faded opacity)
- Update on confirmation
- Rollback on failure

Caching

- Cache metric values (5min)
- Cache chat history (local DB)
- Prefetch likely next screens
- Image thumbnails cached

Animation Performance

- Use transform/opacity only (GPU)
- Avoid layout thrashing
- RequestAnimationFrame for custom animations
- Reduce motion on low-end devices

State Management

View States

```
{
  date: 'today' | ISO date string,
  isHistorical: boolean,
  messages: Message[],
  metrics: {
    chi: MetricData,
    nsl: MetricData,
    exposure: MetricData
  },
  suggestedQuestions: string[],
  quickActions: Action[],
  isLoading: boolean,
  hasError: boolean
}
```

Message Types

```
{
  id: string,
  type: 'twin' | 'user' | 'system',
  content: string,
  timestamp: Date,
  richContent?: {
    type: 'metric' | 'action-grid' | 'image',
    data: any
  },
```

```
  status: 'pending' | 'sent' | 'failed'
}
```

Development Handoff Notes

Component Library

Build these reusable components:

1. `MessageBubble` - Twin/user/system messages
2. `MetricCard` - Compact metric display
3. `QuickActionGrid` - Action button grid
4. `SuggestedQuestion` - Chip button
5. `ChatInput` - Input bar with all features
6. `TypingIndicator` - Animated dots
7. `RichMessage` - Container for embedded content

API Endpoints Needed

- `GET /chat/history?date=YYYY-MM-DD` - Fetch messages
- `GET /metrics/summary?date=YYYY-MM-DD` - Fetch metrics
- `POST /chat/message` - Send message
- `GET /chat/suggestions` - Get suggested questions
- `POST /chat/voice` - Upload voice recording
- `POST /chat/image` - Upload image

State Management

- Consider Zustand or Context API
- Separate stores for: Chat, Metrics, UI
- WebSocket for real-time updates
- Optimistic updates for better UX

Testing Checklist

- [ ] Message rendering (all types)
- [ ] Metric card interactions
- [ ] Quick action functionality
- [ ] Input bar (text, voice, image)
- [ ] Keyboard handling
- [ ] Scroll behavior
- [ ] Date navigation
- [ ] Error states
- [ ] Loading states
- [ ] Accessibility
- [ ] Dark mode (if applicable)
- [ ] Different device sizes

Design Files Deliverables

For Figma handoff:

1. **Component library** with all variants
2. **Full screen designs** for all states
3. **Interactive prototype** with transitions
4. **Animation specs** with timing curves
5. **Responsive layouts** for all breakpoints
6. **Icon set** for all UI elements
7. **Illustration assets** (lotus, charts, etc.)
8. **Dark mode** (if applicable)

Next Steps

1. **Review with team** - Validate approach
2. **Build component library in Figma** - Create all variants
3. **Create interactive prototype** - Show full flow
4. **Developer handoff** - Provide all specs + assets
5. **Iterate based on feedback** - Refine as needed

This specification provides everything needed to build the Today section with pixel-perfect accuracy and production-ready quality.